Karim El Yaagoubi

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Summary

Results-driven Software & Data Engineer with hands-on experience building scalable backend systems, semantic data pipelines, and production mobile applications. Proficient in Java, Spring Boot, and cloud-native tooling. With a track record of shipping open-source libraries and revenue-generating products.

Professional Experience

Astound Technology Jul 2026 – Oct 2026
Backend Software Engineer Intern *Casablanca, Morocco*

- Developed RESTful APIs using **Spring Boot**, following layered architecture and clean coding practices to ensure maintainability and scalability.
- Designed and implemented data models with **MongoDB** and **PostgreSQL**, optimizing schema design for read-heavy workloads.
- Integrated authentication and authorization using **Keycloak** with JWT tokens and role-based access control (RBAC), securing endpoints across microservices.

Projects

Goldfinch | ***C*** Sep 2026 – Present

- Engineered a lightweight RDF triplestore in **C** that stores triples in memory and executes basic SPARQL queries with prefix, variable, IRI, and triple-pattern matching support.
- Implemented a custom lexer and recursive-descent parser for a SPARQL subset, handling query planning and pattern evaluation from scratch.

Semantic Modeling & Analytics of Cancer Data | ***Python, RDF, SPARQL, OWL*** Feb 2026 – Jun 2026

- Designed a custom OWL ontology modeling breast cancer knowledge across 5 classes (Patient, Diagnosis, Treatment, Gene Expression, Mutation).
- Built a Python semantic lifting pipeline that transformed the METABRIC dataset (**1,904 patients** × **693 variables**) into an RDF knowledge graph of ∼**3.85M triples**.
- Benchmarked AllegroGraph, Blazegraph, and Virtuoso in isolated Docker containers on query latency, RAM usage, and disk footprint.

Audio Tag Editor | ***Dart, Java, Flutter, BLoC, Android*** Aug 2025

- Developed and maintained an **open-source Flutter plugin** for reading and editing audio metadata across MP3, M4A, FLAC, OGG, AIFF, WAV, and WMA formats.
- Implemented platform-specific Android code in **Java** and bridged it to Dart via Flutter method channels for efficient native file I/O.
- Built and launched a **production Flutter mobile application** on Google Play using **BLoC architecture** for scalable state management.
- Achieved **10,000+ installs** and **60+ paying users**, generating recurring revenue on Google Play.

CourierFlow | ***Flutter, Spring Boot, GraphHopper, JWT*** May 2025

- Developed a **Flutter mobile application** for delivery management with real-time route tracking and optimized delivery workflows.
- Integrated with a **Spring Boot backend API** leveraging GraphHopper for route optimization, reducing estimated delivery times.
- Implemented **JWT-based authentication** for secure communication between mobile client and backend services.

Certifications

Oracle Certified Professional: Java SE 17 Developer (1Z0-829)

Technical Skills

Languages: Java, Dart, C, Python, PL/SQL
Backend & APIs: Spring Boot, RESTful APIs, JWT, Keycloak
Data & Semantic Web: SPARQL, RDF, OWL, Knowledge Graphs, AllegroGraph, Blazegraph, Virtuoso
Databases: PostgreSQL, MongoDB
Mobile: Flutter, BLoC, Android (Java)
Cloud & DevOps: Docker, Linux, Git, GitHub
Core CS: Data Structures, Algorithms, Object-Oriented Design

Education

National School of Applied Sciences (ENSA) Morocco
Engineering Degree – Data Science and Cloud Computing

Classes Préparatoires aux Grandes Écoles (CPGE) Morocco
Technology and Industrial Science